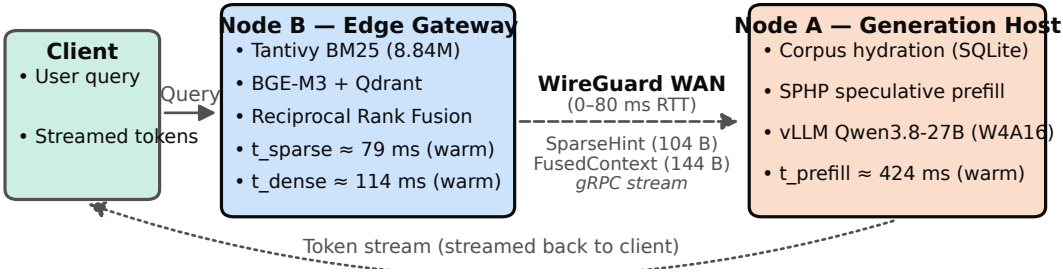
Client
• User query
• Streamed tokens
Query
Node B — Edge Gateway
• Tantivy BM25 (8.84M)
• BGE-M3 + Qdrant
• Reciprocal Rank Fusion
• t_sparse ≈ 79 ms (warm)
• t_dense ≈ 114 ms (warm)
WireGuard WAN
(0–80 ms RTT)
SparseHint (104 B)
FusedContext (144 B)
gRPC stream
Node A — Generation Host
• Corpus hydration (SQLite)
• SPHP speculative prefill
• vLLM Qwen3.8-27B (W4A16)
• t_prefill ≈ 424 ms (warm)
Token stream (streamed back to client)